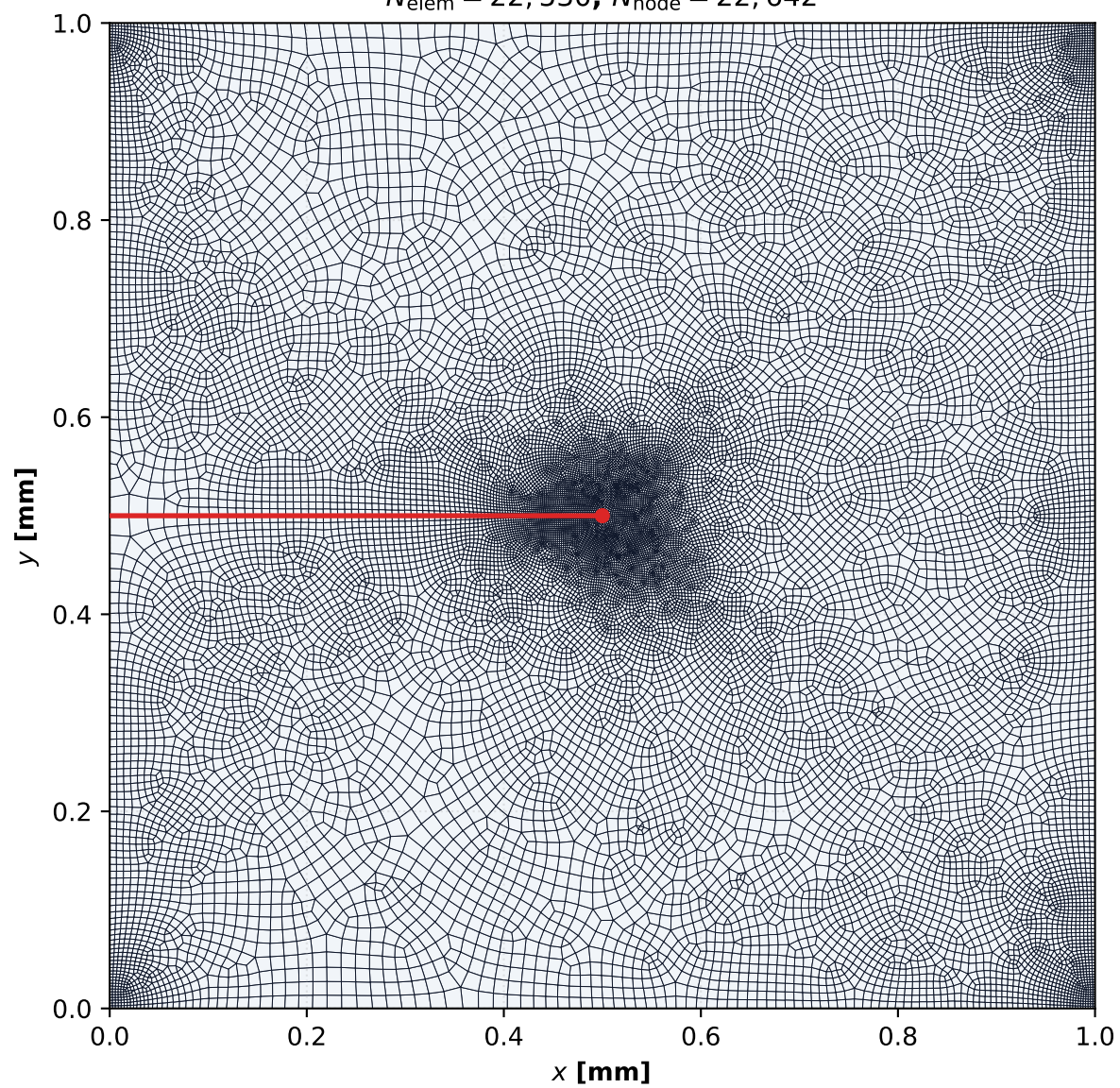


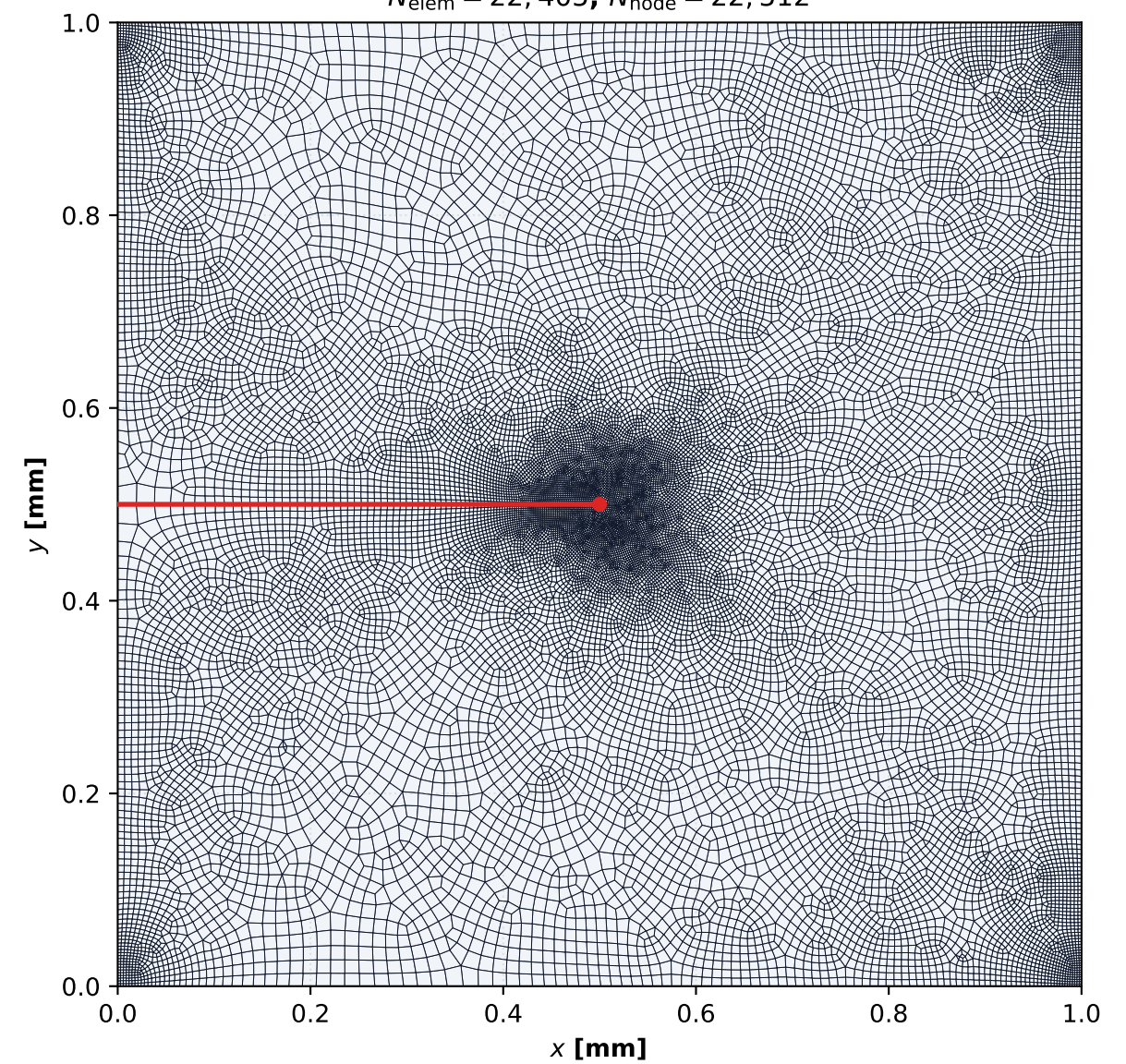
(a) Step-1 Final Frame Mesh ($u_x = 10\ \mu\text{m}$)

$N_{\text{elem}} = 22,530$, $N_{\text{node}} = 22,642$



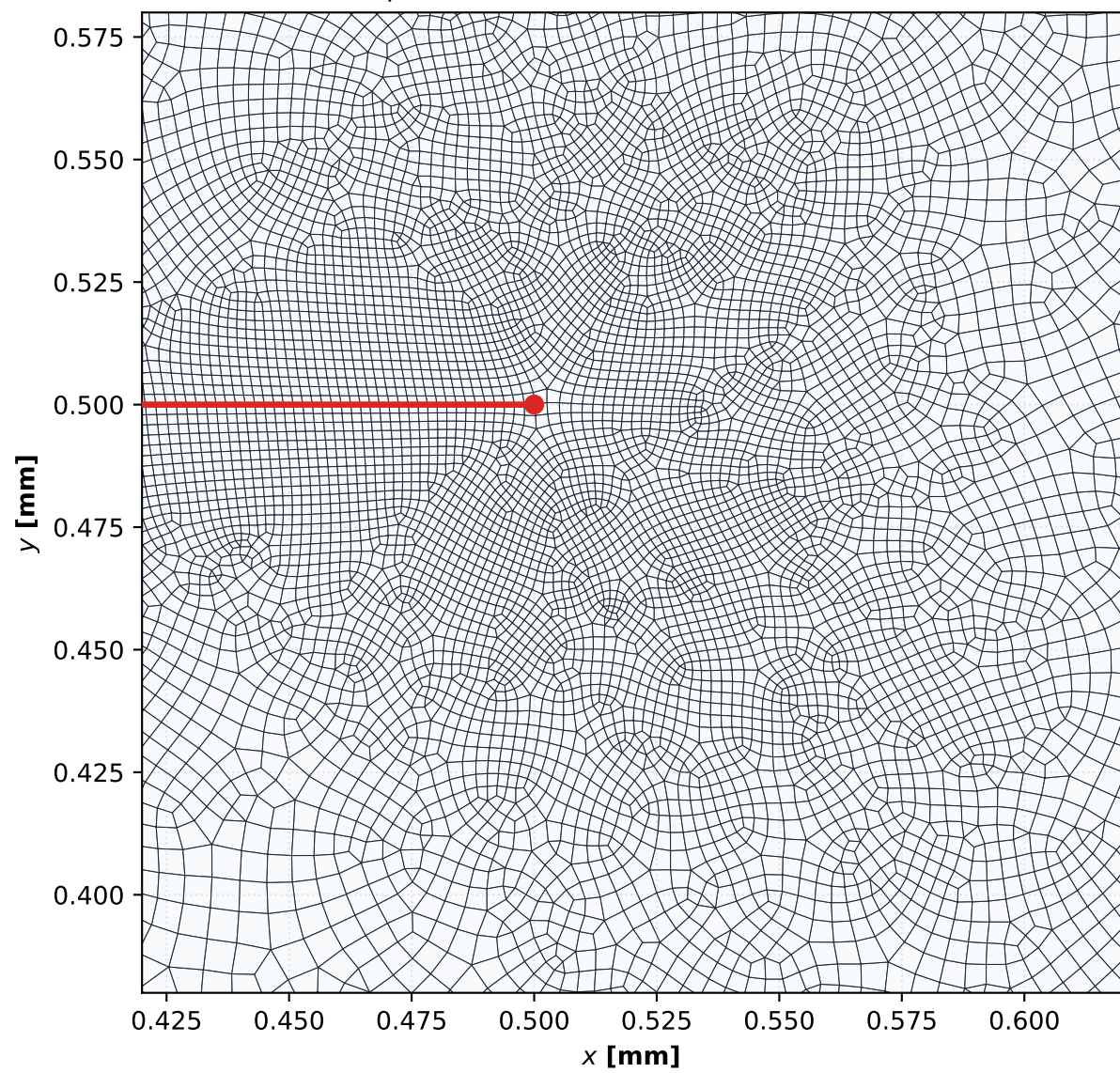
(b) Step-2 Final Frame Mesh ($u_x = 20\ \mu\text{m}$)

$N_{\text{elem}} = 22,405$, $N_{\text{node}} = 22,512$



(c) Step-1 Crack-Tip Singularity Zoom

$h_{\text{min}} = 1.00\ \mu\text{m}$, Refinement Fan Centered on (0.5, 0.5)



(d) Step-2 Crack-Tip Singularity Zoom

$h_{\text{min}} = 1.00\ \mu\text{m}$, Refinement Fan Centered on (0.5, 0.5)

